

This study aims to evaluate the extent to which pre-draft information, particularly NFL Combine performance and draft position, can predict long-term player success in the NFL. Player success is measured using Weighted Approximate Value (AV), a metric developed by Doug Drinen of Pro Football Reference to estimate a player's overall career contribution. Unlike standard Approximate Value, the weighted version places greater emphasis on a player's best seasons while accounting for career longevity, providing a more balanced measure of performance across different career trajectories. The primary objective of this study is to determine whether athletic testing metrics provide meaningful predictive power beyond draft capital, which is often considered a proxy for team expectations and scouting evaluations.

Additionally, this analysis compares multiple modeling approaches, including linear regression, ridge regression, lasso regression, and random forest, to assess which method produces the most accurate predictions. Ultimately, the goal is to better understand the relationship between pre-draft evaluations and actual career outcomes, and to determine whether NFL Combine data can be used to identify potential draft "steals" or overvalued players.

The majority of the data used came from a Kaggle dataset named "NFL Combine Statistics" by Isaak Sipple. This dataset contains data collected at the NFL Combine from years 2000-2025 and features the following data: player, position, school, height, weight, 40-yard dash, vertical jump, bench press, broad jump, 3-cone drill, and the 20-yard shuttle. This dataset was joined with stat tables from Pro Football Reference that included the player's team they were picked by, the round they were picked, their exact pick, the draft year, and their Weighted Career Approximate Value (the target variable).

The data was also cleaned and processed further before being used in analysis. First, all the data prior to 2010 was filtered out due to the shift in the way NFL games are played. Restricting the data to post-2010 seasons helps ensure that the analysis reflects the modern NFL, where player evaluation and on-field performance are more closely aligned with current offensive and defensive trends.

Irrelevant variables such as player name, team, and school were dropped, while missing values in Combine performance metrics were addressed through mean imputation to preserve sample size. Additionally, position was treated as a categorical variable to account for structural differences across roles. To evaluate model performance, a hybrid data splitting strategy was used. First, a time-based split was applied to separate historical data from future predictions, ensuring that the model was trained only on past information. Within the training data, a random partition (80/20) was used to create training and validation sets, allowing for model tuning and performance assessment. A separate dataset containing 2025 draft prospects was retained for final out-of-sample prediction, simulating a real-world forecasting scenario.

To evaluate the relationship between pre-draft characteristics and NFL player success, multiple predictive modeling techniques were applied using Weighted Approximate Value (AV) as the response variable. The analysis began with a baseline linear regression model, which included position, physical attributes, Combine performance metrics, and draft pick as predictors. This approach provided an interpretable benchmark for understanding the direction

and magnitude of relationships between variables and career outcomes. To address potential multicollinearity among Combine metrics, regularization techniques were then implemented. Ridge regression was used to shrink coefficients while retaining all predictors, improving model stability in the presence of correlated features. Lasso regression was also applied to perform variable selection by shrinking weaker predictors toward zero, allowing for identification of the most influential variables. Both models were trained using cross-validation to determine the optimal penalty parameter (λ), ensuring that model complexity was balanced with predictive performance. Finally, a random forest model was estimated to capture potential nonlinear relationships and interactions between predictors that linear models cannot account for. This ensemble method constructs multiple decision trees using bootstrapped samples of the data and aggregates their predictions, reducing variance and improving generalization.

Model performance was evaluated using Root Mean Squared Error (RMSE) as the primary metric and R^2 as a secondary measure of explained variance, both calculated on the validation dataset. This multi-model approach allows for a comprehensive comparison of linear and nonlinear methods in predicting NFL career success.

The performance of each model was evaluated using Root Mean Squared Error (RMSE) and R^2 on the validation dataset. The baseline linear regression model produced an RMSE of 19.88 and an R^2 of 0.239, indicating that approximately 24% of the variance in AV can be explained by the predictors. Ridge and lasso regression yielded nearly identical results, with ridge achieving a slightly lower RMSE of 19.87 and R^2 of 0.240, and lasso producing an RMSE of 19.88 and R^2 of 0.239. These minimal differences suggest that regularization provided little improvement over the standard linear model.

In contrast, the random forest model demonstrated a substantial increase in predictive performance, achieving an RMSE of 17.63 and an R^2 of 0.334. This represents a meaningful reduction in prediction error and an increase in explained variance compared to the linear and regularized models. The improved performance indicates that nonlinear relationships and interactions among predictors play an important role in explaining NFL career success. Variable importance measures from the random forest model highlight draft position, weight, 40-yard dash time, and 3-cone drill performance as the most influential predictors.

Overall, these results suggest that while draft position and player characteristics provide some predictive power, linear models are limited in their ability to capture the complexity of player outcomes. More flexible machine learning approaches, such as random forest, are better suited for modeling these nonlinear patterns.

Several limitations should be considered when interpreting the results of this analysis. First, the dataset contains a substantial amount of missing Combine data, as not all players participate in every drill. Although mean imputation was used to address these missing values, this approach may oversimplify the data and fail to capture meaningful differences between players who did and did not participate in specific tests. Additionally, the missingness itself may not be random, as higher-profile prospects often skip drills, potentially introducing bias into the model.

Second, the use of Weighted Approximate Value as the target variable presents inherent limitations. While AV is designed to provide a comprehensive measure of player performance, it may not fully capture positional differences, contextual factors, or the true impact of a player on team success. As a result, the model's predictions are dependent on a metric that may not perfectly reflect real-world performance. Third, the models rely solely on pre-draft variables such as Combine metrics, physical attributes, and draft position. Important factors that influence NFL success—such as college production, injury history, team fit, coaching quality, and opportunity—are not included in the dataset. The omission of these variables likely contributes to the relatively modest R^2 values observed across all models.

Finally, while the random forest model improved predictive performance, it introduced reduced interpretability compared to linear models. Unlike regression-based approaches, it is more difficult to directly quantify the impact of individual predictors, which may limit the ability to draw clear conclusions about specific variables. Additionally, the analysis assumes that relationships observed in historical data will generalize to future draft classes, which may not fully account for changes in league trends, player development, or scouting practices over time.

The results of this analysis suggest that predictive models such as linear regression, ridge, lasso, and random forest are best utilized as supplemental tools rather than standalone decision-making systems. While these models capture meaningful patterns—particularly the importance of draft position and select athletic metrics—they do not fully account for the complexity of NFL player development. As such, their greatest value lies in serving as an initial screening mechanism to identify players who may be undervalued or overvalued relative to expectations. Model predictions can be used to flag potential “steals” or “risk” prospects, which can then be further evaluated through film analysis, scouting reports, and contextual factors such as team fit and injury history. In this way, predictive modeling provides a strong analytical foundation while complementing, rather than replacing, traditional scouting methods.

In conclusion, this study examined the extent to which pre-draft information, including NFL Combine performance and draft position, can be used to predict long-term player success as measured by AV. The results show that while these variables provide some predictive power, much of the variation in NFL career outcomes remains unexplained. Linear, ridge, and lasso regression models produced nearly identical results, indicating that the underlying relationships are relatively simple and that regularization does not substantially improve performance. However, the random forest model achieved a meaningful improvement in accuracy, suggesting that nonlinear relationships and interactions among variables play an important role in determining player success. Despite this improvement, the overall predictive power of the models remains moderate, highlighting the inherent uncertainty in projecting NFL careers. These findings reinforce the idea that while data-driven approaches can enhance player evaluation, they should be used alongside traditional scouting methods to form a more complete assessment of player potential. These findings also suggest that draft position may already incorporate much of the information available to teams, limiting the additional predictive value of Combine metrics.